

From Like to Unlike: Integrate Preference Revision into Hierarchical Semantic RL-based Music Recommendation

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Abstract

Reinforcement learning-based recommender systems typically treat user feedback as immediate and static, overlooking post-consumption preference revisions such as like-to-unlike transitions. To address this limitation, we propose a Revision-Aware Hierarchical Semantic Actor-Critic (RA-HSAC) framework for sequential music recommendation. Leveraging the industrial-scale Yambda-50M dataset, our approach explicitly models temporal preference drift through two novel mechanisms: Retrospective Revision Credit Assignment, which traces delayed retraction signals to correct historical policy updates, and Revision-Aware Policy Intervention, which dynamically penalizes semantic paths associated with user dissonance. Offline experiments demonstrate that RA-HSAC consistently outperforms state-of-the-art baselines in cumulative reward and interaction depth. By transforming corrective feedback into actionable policy refinements, this work advances RL-based recommendation toward a truly user-centric paradigm that robustly adapts to evolving preferences.

Keywords: Recommender Systems, Preference Revision, Reinforcement Learning, Hierarchical Semantic Representation, Music Recommendation

1. Introduction

Recommender systems (RSs) have become the core infrastructure of digital platforms, shaping user engagement, content distribution, and long-term platform value. In domains such as music streaming and short-video platforms, user-system interactions are inherently sequential, context-dependent, and dynamically evolving. Consequently, Reinforcement Learning (RL) has emerged as a natural paradigm for recommendation, as its mathematical framework explicitly optimizes long-term cumulative utility rather than short-term, myopic accuracy (Zhao et al., 2018). Most recently, to tackle the vast and semantically rich action spaces inherent in real-world recommendations, researchers have introduced Hierarchical Semantic Reinforcement Learning (HSRL) (Wang et al., 2026). Despite these advancements, a fundamental limitation persists: current RL-based recommender systems typically model user feedback as immediate, stable, and unidimensional. In practice, however, user preferences are neither static nor fully revealed at the moment of interaction. A user may initially express positive feedback toward an item (e.g., "like") and subsequently reverse that evaluation (e.g., "unlike"). Such feedback transitions constitute a non-trivial proportion of interactions in real-world platforms (as shown in Figure 1), reflecting a phenomenon of *post-consumption preference revision* that standard recommendation frameworks fail to capture.

Understanding this behavioral drift is especially critical in nearly zero-cost consumption environments like music streaming, where choices carry minimal explicit cost. In such settings, initial feedback often reflects hesitation or tentative exploration rather than a permanent ground truth. Traditional RL paradigms struggle with this because they lack the theoretical mechanisms to process changing preferences over time. The utility gap offers a more precise explanation for this behavior, capturing the divergence between decision utility (anticipated value at the moment of "like") and experienced utility (realized hedonic experience during consumption) (Kahneman et al., 1997). When the experienced utility of a music track fails to meet the predicted utility, users experience post-consumption dissonance (Festinger, 1957). In this context, an "unlike" action serves as a behavioral correction to reduce this dissonance, realigning the user's digital profile with their updated internal state. Furthermore, preferences naturally evolve due to satiation and projection bias (Loewenstein et al., 2003); what a user enjoys in one context may not hold in another. While prior research has explored preference revision in group decision-making driven by social influence (Aribarg et al., 2002), our work focuses on the temporal preference changes that arise from post-consumption evaluation in individual-level sequential decision settings.

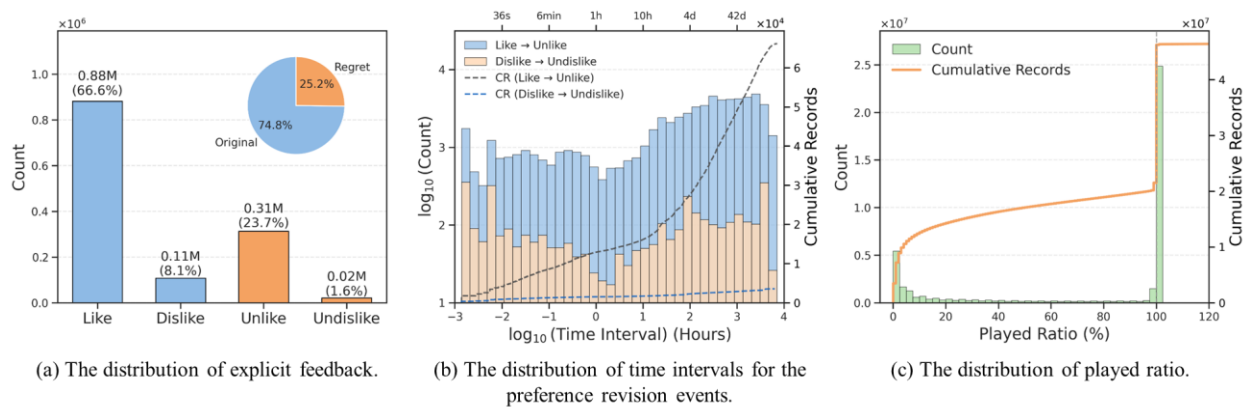


Figure 1: Statistics on user preference revision events in the Yambda-50M dataset.

Neglecting these revision signals discards critical information about preference stability and volatility, both of which are essential for optimizing long-term engagement. The historical absence of this modeling in RSs stems from two primary factors. **First, data limitations:** current studies are constrained by the lack of fine-grained feedback sequences in public datasets. To investigate this challenge, we leverage the industrial-scale Yambda dataset (Ploshkin et al., 2025), which distinctly records explicit positive/negative signals (like, dislike) alongside their subsequent retractions (unlike, undislike). **Second, technical constraints:** existing RL algorithms assume feedback is immediate. While some studies incorporate negative feedback into RL models, dynamic preference revisions cannot be directly equated with simple negative feedback.

Modeling preference revision presents three significant complexities: **(1) Complex Etiology:** Preference revisions arise from intertwined extrinsic and intrinsic factors. Extrinsic factors stem from the system's presented item sequence, which may trigger new associations, while intrinsic factors reflect the endogenous volatility of user preferences (e.g., shifting moods or satiation). **(2) Diverse Manifestations:** Revisions manifest heterogeneously, ranging from engagement-based drops (extremely low played ratios) to explicit feedback retractions in later sessions. **(3) Imbalanced Distribution:** Statistical evidence reveals dual sparsity; explicit feedback is highly

unbalanced (Figure 1a), and the time intervals between an initial action and its revision span multiple orders of magnitude (Figure 1b and 1c).

To address these multifaceted challenges, we propose a novel framework that integrates temporal preference revision into a Hierarchical Semantic Actor-Critic (HSAC) architecture. By operating as a computational system for preference regulation (Zeelenberg & Pieters, 2007), the agent proactively modifies its policy to manage user dissatisfaction. By distinguishing whether a revision is enacted at the current trajectory, our framework categorizes preference revisions into extrinsic and intrinsic types and processes them separately. Consequently, the main contributions are twofold: **(1) Retrospective Revision Credit Assignment:** For extrinsic revisions, we quantify delayed retraction signals into a temporal intensity factor. This factor is traced back to the original decision point, enabling a posterior correction of the advantage function to refine policy updates based on the experienced utility gap. **(2) Revision-Aware Policy Intervention:** For intrinsic revisions, the hierarchical semantic structure facilitates dynamic intervention. By maintaining a memory pool of "failed" semantic paths, a semantic intervention score is introduced that acts as a "soft mask" during inference, steering the agent away from trajectories historically associated with user dissonance. From a system perspective, the goal is not merely to predict, but to adapt policies that are robust to preference drift and capable of incorporating corrective feedback over time.

In summary, this work presents the first systematic analysis of temporal preference revision in sequential music recommendation, transcending the limitations of existing RL methods that treat feedback as static and immediate. By integrating behavioral corrections through retrospective revision credit assignment and proactive policy intervention, we move beyond simple positive-signal maximization toward a truly user-centric paradigm, one that holistically captures the complexities of preference dynamics and learns as effectively from preference withdrawals as it does from initial acquisitions.

2. Methodology

2.1 Problem Formulation

The sequential music recommendation task is modeled as a Markov Decision Process (MDP), denoted by the tuple $\mathcal{M} = \langle \mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{R}, \gamma \rangle$:

State Space ($\mathcal{S}$): At time step t , the state $s_t \in \mathcal{S}$ summarizes the user's context, including their recent interaction history.

Action Space ($\mathcal{A}$): To decouple the policy from a massive and dynamic item catalog, we employ a Semantic Action Space (SAS). An action $a_t \in \mathcal{A}$ is a list of k items. Each item is represented by a sequence of semantic tokens $z = [z_1, \dots, z_L]$, where $z_1 \in \mathcal{V}_\ell$.

Transition and Reward ($\mathcal{P}, \mathcal{R}$): $\mathcal{P}(s_{t+1}|s_t, a_t)$ denotes the state transition, and $r_t = \mathcal{R}(s_t, a_t, s_{t+1})$ is the reward. $\gamma \in (0, 1)$ is the discount factor to balance the reward term.

Policy (π_θ): The policy network generates a semantic action z_t , which is mapped to an effect action a_t via a fixed Codebook: $a_t = \text{Codebook}(z_t) \in \mathcal{A}$.

The raw data employed in this study are structured as sequences of user events, which are misaligned with the time steps corresponding to system actions. To address this discrepancy, we redefine a time step as the timestamp interval between two consecutive changes in the target item within user event sequences. Under this formulation, each step t may encompass one or multiple user events, and without altering the original chronological event order.

2.2 Hierarchical Semantic Actor-Critic

HSAC framework is utilized to model the "coarse-to-fine" dependency of semantic tokens. To fully leverage the advantages of Semantic ID, it implements the Actor and Critic as Hierarchical Policy Network (HPN) and Multi-Level Critic (MLC), respectively (Wang et al., 2026). The HPN encodes the interaction history s_{hist} into an initial global context vector $c_0 = \text{Encoder}(s_{\text{hist}})$. It then generates tokens autoregressively:

$$p(z_l|c_{l-1}) = \text{softmax}(W_l c_{l-1}), \pi_\theta(z|s) = \prod_{l=1}^L p(z_l|c_{l-1}). \quad (1)$$

The W_l is the level- l projection matrix. The MLC provides granular value estimations for each semantic level:

$$V_\phi(s, l) = f_\phi(c_l), l = 0, \dots, L. \quad (2)$$

These per-level values are aggregated using learnable importance weights $\tilde{w}_l$ to compute the weighted state value:

$$\tilde{w}_l = \exp(w_l) / \sum_{j=0}^L \exp(w_j), \hat{V}_\phi(s) = \sum_{l=0}^L \tilde{w}_l V_\phi(s, l). \quad (3)$$

Actor-Critic uses the advantage function A_t as the core signal to update the policy. A_t measures the improvement (or deficiency) of the chosen action relative to the expected outcome of the current state:

$$A_t = \hat{Q}(s_t, a_t) - \hat{V}_\phi(s_t) = r_t + \gamma \hat{V}_{\phi'}(s_{t+1}) - \hat{V}_\phi(s_t), \quad (4)$$

where the $\hat{Q}(s_t, a_t)$ is the action-value function, r_t is the immediate reward. Next, we introduce the attribution models of intrinsic and extrinsic preference revisions based on HSAC.

2.3 Retrospective Revision Credit Assignment

Standard RL models struggle with delayed preference revision signals in music recommendation: when a user interacts with an item but later triggers an *unlike* or *undislike* signal. In response, we introduce a fine-grained retrospective mechanism to correct the credit assignment. In a specific trajectory, the immediate reward r_t is set as a function for user feedback type:

$$r_t = \omega_{\text{listen}} \cdot \text{played_ratio} + \omega_{\text{like}} \cdot \mathbb{I}(y_t = \text{like}) - \omega_{\text{dislike}} \cdot \mathbb{I}(y_t = \text{dislike}), \quad (5)$$

where $\mathbb{I}(\cdot)$ is the indicator function, y_t is the user feedback at time t , $\omega_i \in [0, 1]$ are the weight parameters and $i \in \{\text{listen}, \text{like}, \text{dislike}\}$. y_t represents a composite event, which encompasses all the types of user events that occur in time step t we defined. $\text{played_ratio} \in [0, 1]$ is an important feedback label that models the implicit revisions contained within the implicit action of *listen*. When an explicit revision signal (e.g., *unlike*) occurs at t , we define the revision intensity ψ_t :

$$\psi_t = -\lambda_{\text{unlike}} \cdot \mathbb{I}(y_t = \text{unlike}) + \lambda_{\text{undislike}} \cdot \mathbb{I}(y_t = \text{undislike}), \quad (6)$$

where $\lambda_i \in [0, 1]$ are the weight parameters and $i \in \{\text{unlike}, \text{undislike}\}$. Preference revision is incorporated into the reward function by reconstructing the effective reward:

$$r_t \leftarrow r_t + \gamma^{\Delta t} \cdot \psi_t, \quad (7)$$

where $\Delta t = t - t_{\text{orig}}$, $\gamma \in (0, 1)$ is the discount factor, and t_{orig} is the original decision time of the revision signal. This reconstruction of reward r_t ensures that earlier decisions leading to immediate withdrawal incur a larger penalty. The hyperparameters ω_i and λ_i , which modulate the penalty terms, are set to 1 by default in our subsequent experiments.

2.4 Revision-Aware Policy Intervention

Inspired by research on Controllable Generation and Knowledge Injection, the Revision-Aware Policy Intervention module is designed to directly inject integrated user historical cross-trajectory

revision signals into the hierarchical policy generation process. The module maintains a memory pool $\mathcal{B}_{\text{rev}}$ for each user, storing failed semantic paths and their associated revision strengths:

$$\mathcal{B}_{\text{rev}} = \{(z_{1:L}^j, \Phi_j)\}_{j=1}^N, \quad \Phi_j = \gamma_{tr}^{\Delta tr} \cdot \psi_t^j, \quad (8)$$

where Δtr serves as a generalized temporal measure, denoting the count-based distance between the revision signals and the current trajectory. This memory links the user's preference revision trajectory with the actionable item semantic IDs. During inference at level l at time step t , for the candidate token $k \in \mathcal{V}_l$, calculate the semantic intervention score between the prefix path it occupies and the failed semantic path in the revision semantic memory:

$$D_{z_l=k} = \sum_{(z_j, \Phi_j) \in \mathcal{B}_{\text{rev}}} \text{Sim}(z_{1:l-1} \oplus k, z_{1:l}^j) \cdot \Phi_j, \quad (9)$$

where $\text{Sim}(\cdot)$ is the semantic similarity computation function. Leveraging the hierarchical semantic structure, we assign distinct weights to different levels, which can be either predefined hyperparameters or learnable parameters, to simulate real-world decision logic. Benefiting from the fixed semantic codebook, these operations can be efficiently implemented through one-hot encoding and a series of tensor computations, seamlessly integrating into the context of autoregressive decision generation. Next, the semantic intervention score is applied as a "soft mask" on the original HPN logits before sampling, that is, semantic path intervention:

$$p_\theta(z_l = k | c_{l-1}) = \text{softmax}(W_l c_{l-1} + \eta D_{z_l=k}). \quad (10)$$

2.5 Joint Optimization

According to the standard Actor-Critic architecture, the optimization objective of the entire algorithm is constructed in the following way. For the Critic, value function loss is defined as the expected squared error over the data distribution $\mathcal{D}$:

$$\mathcal{L}_V(\phi) = \mathbb{E}_{s_t, a_t \sim \mathcal{D}} \left[\left(\hat{V}_\phi(s_t) - \hat{Q}(s_t, a_t) \right)^2 \right]. \quad (11)$$

For the Actor, the overall policy loss is defined as:

$$\mathcal{L}_\pi(\theta) = \mathbb{E}_{\mathcal{D}} \left[-A_t \frac{1}{k} \sum_{j=1}^k \sum_{l=1}^L \log p_\theta(z_l | c_{j,l-1}) - \lambda_{en} \mathcal{H}_{en}(p_\theta) \right], \quad (12)$$

where $\mathbb{E}_{\mathcal{D}}[\cdot]$ is denoted as the expectation over $(s_t, a_t, r_t, s_{t+1}, y_t) \sim \mathcal{D}$. $\mathcal{H}_{en}$ is the entropy regularization term with a standard implementation. Finally, the Actor and Critic parameters are jointly optimized by minimizing the combined objective:

$$\min_{\theta, \phi} \mathcal{L}_V(\phi) + \mathcal{L}_\pi(\theta). \quad (13)$$

3. Empirical Evaluation

3.1 Experimental setup

Dataset. Yambda is a large-scale multi-modal music recommendation dataset with recommendation interactions and audio embeddings. Our experiments are conducted on the original Yambda-50M (Ploshkin et al., 2025). *Compared Methods.* To evaluate the effectiveness of our proposed framework, we compare our model against the following baselines: SASRec, a Transformer-based sequential recommendation model using supervised learning (Kang and McAuley, 2018); HAC, a hierarchical method for RL-based recommendation (Zhao et al., 2018); and HSRL, which introduces Semantic Action Space for RL-based recommendation (Wang et al., 2026). *Evaluation Metrics.* Two standard metrics commonly used on offline recommendation datasets, *Total Reward* and *Depth*, are employed to measure performance. *Implementation Details.* For Semantic ID construction, we apply RQ-k-means on music embeddings, with SASRec as the actor backbone for autoregressive action generation, consistent with Wang et al. (2026).

3.2 Experimental results

The results of comparative experiments and ablation studies are collectively presented in Table 1. RA-HSAC denotes our Revision-Aware HSAC method. The Δ columns indicate each baseline's percentage performance drop relative to RA-HSAC on each metric.

	Model	Total Reward	Δ (%)	Depth	Δ (%)
comparisons	SASRec	3.35	-49.4	4.12	-51.7
	HAC	5.96	-20.1	7.03	-17.6
	HSRL	<u>6.62</u>	-11.2	<u>7.99</u>	-6.3
ours	RA-HSAC	7.46	--	8.53	--
ablations	w/o RRCA	6.83	-8.4	8.15	-4.5
	w/o RAPI	<u>7.28</u>	-2.4	<u>8.44</u>	-1.1

Table 1: Offline performance comparison ablation study results on Yambda-50M.

Comparative results show that RA-HSAC achieves consistent improvements over baseline models. This enhancement stems from two factors: the powerful sequential decision-making capability and high scalability of hierarchical semantic reinforcement learning, and our fine-grained approach to handling user preference revision signals. Ablation studies demonstrate that both components contribute indispensably to modeling user preference revision. Overall, the results indicate that fine-grained credit assignment and policy intervention enhance HSAC's capability to capture users' dynamically evolving preferences.

4. Conclusion and Future Work

This work addresses the challenge of insufficient modeling of temporal preference revision and post-consumption dissonance in music recommendation by proposing a revision-aware method based on the HSAC framework. Designed to achieve fine-grained retrospective revision credit assignment and proactive policy intervention, our approach dynamically adapts to evolving user preferences. In the future, we aim to extend this framework to more general RL-based recommendation tasks and evaluate its stability and scalability under more stringent real-world conditions.

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